

KSO does SIMON & GARFUNKEL

FOH CHANNEL EQ SETUP — Channels 1-23

Console: Behringer Wing — 6-band parametric channel EQ + LC/HC filters

Venue: Greaves Concert Hall — Northern Kentucky University (637 seats, hardwood floor, adjustable acoustic shell)

Style: Simon & Garfunkel — Vocal-forward / Acoustic-driven / Brushwork drums / Warm strings

EQ Approach: Restrained, supportive, never aggressive | Whole-dB gain values | No high shelf bands

SHOW STYLE — Why This EQ Approach: Simon & Garfunkel material is defined by vocal harmony, fingerpicked acoustic guitar, gentle drum kit (often brushwork), supportive bass, warm strings, and rare jazz-tinged horn arrangements. The defining feature: **nothing is aggressive**. No scooped kick. No bright snare smash. No barking brass. Every channel's EQ aims for *supportive warmth and clarity* rather than modern rock or pop aggression.

VENUE — Greaves Concert Hall: 637-seat acoustically-tuned concert hall with hardwood floor and adjustable acoustic shell. Designed for orchestral and chamber music — warm wood-finished interior with moderate reverberance. Estimated RT60 ~1.5-1.9s. Significantly drier than Memorial Hall but not a black box. Stage has Steinway and Baldwin 9-ft grand pianos.

CONSOLE — Behringer Wing EQ Structure: 6 fully-parametric bands per channel (L, 1, 2, 3, 4, H) plus independent LC and HC filters. L and H bands switchable between Bell and Shelf — these documents specify which is in use. LC slopes: 6/12/18/24 dB/oct. HC slopes: 6/12 dB/oct. All bands sweepable across full audio range with ± 18 dB gain and Q range 0.3-10.

FORMATTING NOTES: Per show standard — whole-dB gain values only, no high shelf bands in use, compression settings not included. Q values express the Wing's standard Q parameter.

INPUT LIST — Channels 1-23

Ch	Channel Name	Mic / Source	Section
1	Kick	Earthworks DM6 SeisMic	DRUMS
2	Snare	Earthworks DM17	DRUMS
3	Hat	Earthworks SR20 Gen 2	DRUMS
4	Overhead	Earthworks SR20 Gen 2	DRUMS
5	Aux Perc	sE Electronics sE8	DRUMS
6	Bass DI	Radial J48	BASS
7	Electric Guitar	Neve RNDI	GUITARS
8	Acoustic Guitar	Neve RNDI	GUITARS
9	Keys	Neve RNDI	GUITARS
10	Piano Low	DPA 4099	PIANO
11	Piano High	DPA 4099	PIANO
12	Violin 1	Countryman B3	STRINGS
13	Violin 2	Countryman B3	STRINGS
14	Violin 3	Countryman B3	STRINGS

Ch	Channel Name	Mic / Source	Section
15	Violin 4	Countryman B3	STRINGS
16	Violin 5	Countryman B3	STRINGS
17	Violin 6	Countryman B3	STRINGS
18	Cello 1	Countryman B3	STRINGS
19	Cello 2	Countryman B3	STRINGS
20	Horns Left (C422 XY)	AKG C422 (stereo, XY)	HORNS
21	Horns Right (C422 XY)	AKG C422 (stereo, XY)	HORNS
22	Flute / Piccolo	TBD — confirm before show	TBD
23	Trombone (Bone)	Warm Audio U87 Jr	TROMBONE

Ch 1 | Kick

Mic: Earthworks DM6 SeisMic

Mic Notes: DM6 is an extended-range dynamic kick mic with usable response down to 20Hz. Captures sub content faithfully without the typical scooped kick sound. For S&G; style, you don't want a scooped modern rock kick — you want a fuller, rounded, supportive thump.

Band	Freq	Type	Gain	Q	Notes
LC	40 Hz	LC	—	18 dB/oct	Engaged. DM6 captures usable down to 20Hz; cut anything below the kick fundamental
L	80 Hz	Shelf	+2 dB	—	Gentle shelf for body — supportive, not boomy
1	250 Hz	Bell	-3 dB	1.2	Cut box/cardboard. S&G; material doesn't need a thick kick
2	800 Hz	Bell	-2 dB	1.5	Light cut for clarity — minimal mid muddiness
3	3.5 kHz	Bell	+2 dB	1.8	Soft beater click for definition without aggression
4	—	OFF	—	—	Not needed
H	—	OFF	—	—	No high shelf — keep S&G; kick warm
HC	8 kHz	HC	—	12 dB/oct	Engaged. Roll off cymbal bleed and high-end rumble

Engineer Notes: Soft, warm, supportive kick. The DM6's natural extended low-frequency character does the work — minimal EQ needed. The 250Hz mud cut keeps the kick clean in Greaves' wood-shell acoustic. No high shelf — S&G; kicks should never sound aggressive or modern.

Ch 2 | Snare

Mic: Earthworks DM17

Mic Notes: DM17 is Earthworks' premium snare/tom dynamic — extremely fast transient response, more like a condenser than a typical snare dynamic. Picks up brushwork and ghost notes that an SM57 would lose entirely. Ideal for S&G; style — captures soft playing with detail.

Band	Freq	Type	Gain	Q	Notes
LC	80 Hz	LC	—	18 dB/oct	Engaged. Remove kick bleed and stage rumble
L	—	OFF	—	—	Snare doesn't need low shelf
1	180 Hz	Bell	+2 dB	1.5	Body and weight — keep snare warm and supportive
2	500 Hz	Bell	-3 dB	1.4	Cut paper/cardboard ring — cleans up snare attack
3	3 kHz	Bell	+2 dB	1.8	Subtle stick attack — restrained for the style
4	8 kHz	Bell	+2 dB	1.5	Brushwork detail and snare wire shimmer — critical for S&G;
H	—	OFF	—	—	Band 4 covers air
HC	—	OFF	—	—	Open top end for brushwork detail

Engineer Notes: Snare is critical for S&G; material — brushwork is the defining drum technique. Band 4 at 8kHz is the most important boost on this channel. The DM17's fast transient response captures the brush detail that the EQ then enhances.

Ch 3 | Hat

Mic: Earthworks SR20 Gen 2

Mic Notes: SR20 Gen 2 is a small-diaphragm pencil condenser, cardioid, ruler-flat response, very low self-noise. On hat it gives accurate stick definition without the harshness of cheaper condensers. Boom mounted.

Band	Freq	Type	Gain	Q	Notes
LC	300 Hz	LC	—	24 dB/oct	Aggressive HPF. Hat doesn't need anything below 300Hz; reject snare/kick bleed
L	—	OFF	—	—	—
1	800 Hz	Bell	-2 dB	1.2	Cut clank/wash from brass body
2	2 kHz	Bell	-2 dB	1.5	Gentle cut for harshness — keep hat polite
3	—	OFF	—	—	—
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve cymbal sparkle

Engineer Notes: Aggressive 300Hz HPF rejects bleed; subtle midrange cuts keep hat polite. Open top end preserves sparkle — for S&G; style hat should be present but never obtrusive.

Ch 4 | Overhead

Mic: Earthworks SR20 Gen 2

Mic Notes: Single OH (mono), boom mounted. With one overhead this is the kit's blend — captures cymbals and overall drum image. SR20's flat response gives a natural representation. For S&G; style OH should sound like the kit in a natural acoustic.

Band	Freq	Type	Gain	Q	Notes
LC	150 Hz	LC	—	18 dB/oct	Engaged. Reject low-end mud from kick and stage
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.5	Slight cut to clean up boxiness — Greaves' warm acoustic plus mono OH stacks here
2	2 kHz	Bell	-2 dB	1.5	Keep upper mids gentle — restraint over aggression
3	6 kHz	Bell	+2 dB	1.8	Cymbal definition without harshness
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve cymbal extension

Engineer Notes: Mono OH carries kit blend. The 150Hz HPF and 400Hz cut keep it from competing with the kick and snare in the low-mid. 6kHz boost gives cymbal sparkle without harshness.

Ch 5 | Aux Perc

Mic: sE Electronics sE8

Mic Notes: sE8 is a small-diaphragm cardioid condenser. Boom-mounted on aux percussion. Specific instrument(s) TBD — settings are permissive for shaker/tambourine/hand drums. Adjust based on actual instrument played during soundcheck.

Band	Freq	Type	Gain	Q	Notes
LC	150 Hz	LC	—	18 dB/oct	Engaged. Aux perc doesn't need low-end
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.5	Light cut for transparency in mix
2	—	OFF	—	—	Adjust based on actual instrument
3	5 kHz	Bell	+2 dB	1.8	Transient definition — shakers, tambourine, hand drums all benefit
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Open top for transient detail

Engineer Notes: Permissive starting point. Confirm specific percussion instrument at soundcheck and adjust Band 2 if needed (e.g., congas may want a 200Hz boost; tambourine may need a 4kHz cut for harshness).

Ch 6 | Bass DI

Mic: Radial J48

Mic Notes: J48 is a clean, transparent active DI with phantom power — Radial's industry-standard live bass DI. No transformer coloration. Delivers bass exactly as it is — EQ has to do all character work. For S&G; style think the warm fingerstyle bass on 'The Boxer' or walking lines on 'Cecilia'.

Band	Freq	Type	Gain	Q	Notes
LC	40 Hz	LC	—	18 dB/oct	Engaged. Cut sub rumble below usable bass
L	80 Hz	Shelf	+2 dB	—	Body and warmth
1	250 Hz	Bell	-3 dB	1.2	Cut mud — Greaves' wood acoustic stacks here
2	800 Hz	Bell	+2 dB	1.5	Note definition for fingerstyle clarity
3	2 kHz	Bell	+1 dB	1.8	Subtle pick/string articulation
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	10 kHz	HC	—	12 dB/oct	Engaged. No useful content above 10kHz from a clean bass DI

Engineer Notes: Clean DI means EQ does all the work. Low shelf adds warmth, 250Hz mud cut keeps it tight, 800Hz boost gives note definition. The 2kHz subtle boost adds articulation for fingerstyle playing without making the bass clicky.

Ch 7 | Electric Guitar

Mic: Neve RNDI

Mic Notes: RNDI direct from electric guitar. Likely capturing clean Strat or hollowbody for S&G; material. Without amp/cab simulation, this signal is clean and direct. For S&G; style think 'Mrs. Robinson' clean tones — round, warm, never aggressive. RNDI's transformer adds gentle harmonic richness so EQ doesn't need to add character.

Band	Freq	Type	Gain	Q	Notes
LC	80 Hz	LC	—	18 dB/oct	Engaged. Electric guitar has nothing useful below 80Hz
L	—	OFF	—	—	—
1	200 Hz	Bell	+2 dB	1.0	Body warmth — keep DI guitar from sounding thin
2	500 Hz	Bell	-2 dB	1.4	Cut DI honkiness
3	2 kHz	Bell	+2 dB	1.5	Presence and pick attack
4	5 kHz	Bell	-2 dB	1.8	Tame pick harshness — common DI complaint
H	—	OFF	—	—	No high shelf per spec
HC	8 kHz	HC	—	12 dB/oct	Engaged. Cab simulation from EQ — DI without amp sim has unwanted content above 8kHz

Engineer Notes: DI without cab sim needs significant EQ to feel like a real guitar. Bands 1 and 4 do the cab simulation — Band 1 restores body, Band 4 tames the pick harshness that an amp would naturally roll off. The 8kHz HC supplements this.

Ch 8 | Acoustic Guitar

Mic: Neve RNDI

Mic Notes: RNDI on acoustic guitar via piezo pickup. For S&G; style the acoustic guitar IS the sound — fingerpicked Martin tones drive every signature song. One of the most important channels in the show. Piezo DIs lose body resonance and exaggerate the bridge area. RNDI's transformer warms the DI but cannot fix piezo character entirely.

Band	Freq	Type	Gain	Q	Notes
LC	100 Hz	LC	—	18 dB/oct	Engaged. Lowest open string E2 = 82Hz; HPF at 100Hz
L	200 Hz	Shelf	+2 dB	—	Body restoration — piezo loses body cavity resonance
1	400 Hz	Bell	-3 dB	1.4	Cut piezo box resonance
2	2 kHz	Bell	-3 dB	1.5	The classic piezo 'quack' zone — most important cut on this channel
3	5 kHz	Bell	+2 dB	1.8	Pick definition and string clarity for fingerpicking
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve fingerpicking sparkle

Engineer Notes: Most critical channel for S&G; material. The piezo quack at 2kHz must be cut hard — this is where the harshness lives. Body restoration at 200Hz and pick clarity at 5kHz transform the DI into something that sounds like a real Martin. Open top end for fingerpicking sparkle.

Ch 9 | Keys

Mic: Neve RNDI

Mic Notes: RNDI from keyboard — likely Nord, Roland, or similar stage keys covering piano, organ, EP patches. Signal is already-mixed stereo (this is mono Ch 9 only, possibly summed or one side). RNDI's transformer works well on keys per manufacturer testimonials. EQ should be permissive — let the patch through.

Band	Freq	Type	Gain	Q	Notes
LC	50 Hz	LC	—	18 dB/oct	Engaged. Cut sub content; keys patches go low but rumble is undesirable
L	—	OFF	—	—	Keys patch already shaped — don't add low-end
1	300 Hz	Bell	-2 dB	1.2	Light mud cut for mix clarity
2	—	OFF	—	—	—
3	3 kHz	Bell	+1 dB	1.5	Subtle presence lift; reduce to 0 if keys patch is already bright
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	15 kHz	HC	—	12 dB/oct	Engaged. Cut very high content

Engineer Notes: Permissive EQ — the patch is already shaped at the keyboard. Light mud cut and subtle presence lift only. Adjust Band 3 by ear during soundcheck — bright patches need 0dB, dark patches may want +2dB.

Ch 10 | Piano Low

Mic: DPA 4099

Mic Notes: DPA 4099 with piano clip — supercardioid clip-on, professional capture. Greaves has Steinway and Baldwin 9' grands on stage. For S&G; material piano is essential — 'Bridge Over Troubled Water' lives or dies on piano sound. 4099 inside lid near low strings gives focused capture of bass-register hammers.

Band	Freq	Type	Gain	Q	Notes
LC	60 Hz	LC	—	18 dB/oct	Engaged. Piano lowest A0 = 27.5Hz, but live mix doesn't need below 60Hz
L	100 Hz	Shelf	+2 dB	—	Low register weight — fundamental of low piano notes
1	300 Hz	Bell	-3 dB	1.4	Cut mud — DPA 4099 inside lid picks up boxiness
2	800 Hz	Bell	-2 dB	1.5	Reduce woody resonance buildup
3	3 kHz	Bell	+2 dB	1.5	Hammer attack and note definition for low register
4	—	OFF	—	—	Hi piano channel covers high content
H	—	OFF	—	—	No high shelf per spec
HC	8 kHz	HC	—	12 dB/oct	Engaged. Hi piano channel handles high content; LPF here keeps mics complementary

Engineer Notes: Half of stereo piano pair. Pan left (10 o'clock). Aggressive mid cuts at 300Hz and 800Hz tame the inside-lid boxiness. The 8kHz LPF lets the Hi piano mic carry high content — keep the two mics complementary, not redundant.

Ch 11 | Piano High

Mic: DPA 4099

Mic Notes: DPA 4099 placed near high strings/treble register inside lid. Pairs with Ch 10 — half of stereo piano pair. EQ complements Ch 10, not duplicates it — let each mic do its frequency range job and they sum well in stereo.

Band	Freq	Type	Gain	Q	Notes
LC	150 Hz	LC	—	18 dB/oct	Engaged — aggressive HPF. Hi mic doesn't need lows; let Lo mic carry low end
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Light mud control
2	—	OFF	—	—	—
3	5 kHz	Bell	+2 dB	1.8	High register clarity and brilliance
4	10 kHz	Bell	+2 dB	1.5	Air and shimmer for high notes — critical for S&G; piano sparkle
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve piano harmonic extension

Engineer Notes: Pan right (2 o'clock) to balance the Lo mic. Band 4 at 10kHz is the most important boost — gives the piano its sparkle for S&G; material. The aggressive 150Hz HPF prevents this mic from competing with the Lo mic in the low-end.

Ch 12 | Violin 1

Mic: Countryman B3

Mic Notes: B3 omni clip-mounted on violin near bridge or chinrest. Omni lavalier captures body resonance well but is highly susceptible to bleed and feedback in the 637-seat hall. Violin G3 = 196Hz lowest open string; harmonics extend past 10kHz. Classic violin harshness zone 2-4kHz needs management with 6 violins stacked.

Band	Freq	Type	Gain	Q	Notes
LC	180 Hz	LC	—	18 dB/oct	Engaged. Aggressive HPF — violin lowest is 196Hz; HPF at 180Hz removes body rumble, instrument handling, foot stomps
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Cut box resonance from clip-mount on body
2	3 kHz	Bell	-3 dB	1.5	Tame violin harshness zone — critical with 6 violins stacked
3	6 kHz	Bell	+2 dB	1.8	Bow articulation and string definition
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve violin harmonic extension

Engineer Notes: Violin 1 of 6 — apply identical EQ across all violins for matched section sound. The 3kHz cut is the most critical move — 6 violins stacked at this frequency in Greaves' wood-shell acoustic will sound harsh without it. If a soloist passage emerges, push fader rather than changing EQ.

Ch 13 | Violin 2

Mic: Countryman B3

Mic Notes: B3 omni clip-mounted on violin near bridge or chinrest. Omni lavalier captures body resonance well but is highly susceptible to bleed and feedback in the 637-seat hall. Violin G3 = 196Hz lowest open string; harmonics extend past 10kHz. Classic violin harshness zone 2-4kHz needs management with 6 violins stacked.

Band	Freq	Type	Gain	Q	Notes
LC	180 Hz	LC	—	18 dB/oct	Engaged. Aggressive HPF — violin lowest is 196Hz; HPF at 180Hz removes body rumble, instrument handling, foot stomps
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Cut box resonance from clip-mount on body
2	3 kHz	Bell	-3 dB	1.5	Tame violin harshness zone — critical with 6 violins stacked
3	6 kHz	Bell	+2 dB	1.8	Bow articulation and string definition
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve violin harmonic extension

Engineer Notes: Violin 2 of 6 — apply identical EQ across all violins for matched section sound. The 3kHz cut is the most critical move — 6 violins stacked at this frequency in Greaves' wood-shell acoustic will sound harsh without it. If a soloist passage emerges, push fader rather than changing EQ.

Ch 14 | Violin 3

Mic: Countryman B3

Mic Notes: B3 omni clip-mounted on violin near bridge or chinrest. Omni lavalier captures body resonance well but is highly susceptible to bleed and feedback in the 637-seat hall. Violin G3 = 196Hz lowest open string; harmonics extend past 10kHz. Classic violin harshness zone 2-4kHz needs management with 6 violins stacked.

Band	Freq	Type	Gain	Q	Notes
LC	180 Hz	LC	—	18 dB/oct	Engaged. Aggressive HPF — violin lowest is 196Hz; HPF at 180Hz removes body rumble, instrument handling, foot stomps
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Cut box resonance from clip-mount on body
2	3 kHz	Bell	-3 dB	1.5	Tame violin harshness zone — critical with 6 violins stacked
3	6 kHz	Bell	+2 dB	1.8	Bow articulation and string definition
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve violin harmonic extension

Engineer Notes: Violin 3 of 6 — apply identical EQ across all violins for matched section sound. The 3kHz cut is the most critical move — 6 violins stacked at this frequency in Greaves' wood-shell acoustic will sound harsh without it. If a soloist passage emerges, push fader rather than changing EQ.

Ch 15 | Violin 4

Mic: Countryman B3

Mic Notes: B3 omni clip-mounted on violin near bridge or chinrest. Omni lavalier captures body resonance well but is highly susceptible to bleed and feedback in the 637-seat hall. Violin G3 = 196Hz lowest open string; harmonics extend past 10kHz. Classic violin harshness zone 2-4kHz needs management with 6 violins stacked.

Band	Freq	Type	Gain	Q	Notes
LC	180 Hz	LC	—	18 dB/oct	Engaged. Aggressive HPF — violin lowest is 196Hz; HPF at 180Hz removes body rumble, instrument handling, foot stomps
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Cut box resonance from clip-mount on body
2	3 kHz	Bell	-3 dB	1.5	Tame violin harshness zone — critical with 6 violins stacked
3	6 kHz	Bell	+2 dB	1.8	Bow articulation and string definition
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve violin harmonic extension

Engineer Notes: Violin 4 of 6 — apply identical EQ across all violins for matched section sound. The 3kHz cut is the most critical move — 6 violins stacked at this frequency in Greaves' wood-shell acoustic will sound harsh without it. If a soloist passage emerges, push fader rather than changing EQ.

Ch 16 | Violin 5

Mic: Countryman B3

Mic Notes: B3 omni clip-mounted on violin near bridge or chinrest. Omni lavalier captures body resonance well but is highly susceptible to bleed and feedback in the 637-seat hall. Violin G3 = 196Hz lowest open string; harmonics extend past 10kHz. Classic violin harshness zone 2-4kHz needs management with 6 violins stacked.

Band	Freq	Type	Gain	Q	Notes
LC	180 Hz	LC	—	18 dB/oct	Engaged. Aggressive HPF — violin lowest is 196Hz; HPF at 180Hz removes body rumble, instrument handling, foot stomps
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Cut box resonance from clip-mount on body
2	3 kHz	Bell	-3 dB	1.5	Tame violin harshness zone — critical with 6 violins stacked
3	6 kHz	Bell	+2 dB	1.8	Bow articulation and string definition
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve violin harmonic extension

Engineer Notes: Violin 5 of 6 — apply identical EQ across all violins for matched section sound. The 3kHz cut is the most critical move — 6 violins stacked at this frequency in Greaves' wood-shell acoustic will sound harsh without it. If a soloist passage emerges, push fader rather than changing EQ.

Ch 17 | Violin 6

Mic: Countryman B3

Mic Notes: B3 omni clip-mounted on violin near bridge or chinrest. Omni lavalier captures body resonance well but is highly susceptible to bleed and feedback in the 637-seat hall. Violin G3 = 196Hz lowest open string; harmonics extend past 10kHz. Classic violin harshness zone 2-4kHz needs management with 6 violins stacked.

Band	Freq	Type	Gain	Q	Notes
LC	180 Hz	LC	—	18 dB/oct	Engaged. Aggressive HPF — violin lowest is 196Hz; HPF at 180Hz removes body rumble, instrument handling, foot stomps
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.4	Cut box resonance from clip-mount on body
2	3 kHz	Bell	-3 dB	1.5	Tame violin harshness zone — critical with 6 violins stacked
3	6 kHz	Bell	+2 dB	1.8	Bow articulation and string definition
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve violin harmonic extension

Engineer Notes: Violin 6 of 6 — apply identical EQ across all violins for matched section sound. The 3kHz cut is the most critical move — 6 violins stacked at this frequency in Greaves' wood-shell acoustic will sound harsh without it. If a soloist passage emerges, push fader rather than changing EQ.

Ch 18 | Cello 1

Mic: Countryman B3

Mic Notes: B3 clip-mounted on cello (likely bridge or near f-hole). Cello C2 lowest = 65Hz, fundamentals through 1kHz with harmonics beyond. B3's omni pattern works well for cello body radiation since cellos radiate omnidirectionally. For S&G; context cello provides warmth and depth — think the warm cello in 'El Condor Pasa' arrangements.

Band	Freq	Type	Gain	Q	Notes
LC	60 Hz	LC	—	18 dB/oct	Engaged. Cello low C = 65Hz; HPF at 60Hz preserves fundamental
L	120 Hz	Shelf	+2 dB	—	Body warmth and weight
1	300 Hz	Bell	-2 dB	1.4	Cut woody mud from clip-mount body resonance
2	800 Hz	Bell	+1 dB	1.5	Subtle midrange body for cello character
3	3 kHz	Bell	+2 dB	1.8	Bow attack and presence
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	10 kHz	HC	—	12 dB/oct	Engaged. Cello upper harmonics roll off naturally; LPF reduces room/section bleed

Engineer Notes: Cello 1 of 2 — apply identical EQ to both cellos. The low shelf at 120Hz adds warmth that the omni B3 captures naturally. 10kHz LPF is critical — reduces bleed pickup from the 6 violins and the room without removing cello character.

Ch 19 | Cello 2

Mic: Countryman B3

Mic Notes: B3 clip-mounted on cello (likely bridge or near f-hole). Cello C2 lowest = 65Hz, fundamentals through 1kHz with harmonics beyond. B3's omni pattern works well for cello body radiation since cellos radiate omnidirectionally. For S&G; context cello provides warmth and depth — think the warm cello in 'El Condor Pasa' arrangements.

Band	Freq	Type	Gain	Q	Notes
LC	60 Hz	LC	—	18 dB/oct	Engaged. Cello low C = 65Hz; HPF at 60Hz preserves fundamental
L	120 Hz	Shelf	+2 dB	—	Body warmth and weight
1	300 Hz	Bell	-2 dB	1.4	Cut woody mud from clip-mount body resonance
2	800 Hz	Bell	+1 dB	1.5	Subtle midrange body for cello character
3	3 kHz	Bell	+2 dB	1.8	Bow attack and presence
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	10 kHz	HC	—	12 dB/oct	Engaged. Cello upper harmonics roll off naturally; LPF reduces room/section bleed

Engineer Notes: Cello 2 of 2 — apply identical EQ to both cellos. The low shelf at 120Hz adds warmth that the omni B3 captures naturally. 10kHz LPF is critical — reduces bleed pickup from the 6 violins and the room without removing cello character.

■ HORNS — C422 STEREO PAIR ■

Ch 20 | Horns Left (C422 XY)

Mic: AKG C422 (stereo, XY)

Mic Notes: Single AKG C422 stereo mic in XY mode. Top capsule rotated 45 degrees, both capsules cardioid, capturing horns Left/Right. This is the Left capsule. The C422's CK12 capsules are smoother/fuller than a C414. For S&G; horns think gentle and supportive — 'Mrs. Robinson' style horn parts. PAN: Hard Left.

Band	Freq	Type	Gain	Q	Notes
LC	80 Hz	LC	—	18 dB/oct	Engaged. Cut stage rumble below brass fundamentals
L	—	OFF	—	—	—
1	300 Hz	Bell	-2 dB	1.4	Cut boxy resonance — common at distance from brass
2	1 kHz	Bell	-2 dB	1.5	Tame midrange bark — keeps brass section warm not aggressive
3	5 kHz	Bell	+2 dB	1.8	Brass bite for clarity in mix
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	12 kHz	HC	—	12 dB/oct	Engaged. C422 has extended top end; LPF prevents harsh shimmer in Greaves

Engineer Notes: Apply identical EQ to Ch 20 and Ch 21 for proper XY stereo image. Pan hard left. The C422 in XY captures horn section as natural stereo — the EQ keeps it warm and supportive rather than aggressive.

Ch 21 | Horns Right (C422 XY)

Mic: AKG C422 (stereo, XY)

Mic Notes: Single AKG C422 stereo mic in XY mode. Top capsule rotated 45 degrees, both capsules cardioid, capturing horns Left/Right. This is the Right capsule. The C422's CK12 capsules are smoother/fuller than a C414. For S&G; horns think gentle and supportive — 'Mrs. Robinson' style horn parts. PAN: Hard Right.

Band	Freq	Type	Gain	Q	Notes
LC	80 Hz	LC	—	18 dB/oct	Engaged. Cut stage rumble below brass fundamentals
L	—	OFF	—	—	—
1	300 Hz	Bell	-2 dB	1.4	Cut boxy resonance — common at distance from brass
2	1 kHz	Bell	-2 dB	1.5	Tame midrange bark — keeps brass section warm not aggressive
3	5 kHz	Bell	+2 dB	1.8	Brass bite for clarity in mix
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	12 kHz	HC	—	12 dB/oct	Engaged. C422 has extended top end; LPF prevents harsh shimmer in Greaves

Engineer Notes: Apply identical EQ to Ch 20 and Ch 21 for proper XY stereo image. Pan hard right. The C422 in XY captures horn section as natural stereo — the EQ keeps it warm and supportive rather than aggressive.

Ch 22 | Flute / Piccolo

Mic: TBD — confirm before show

Mic Notes: Mic not specified on input list. Likely candidates from your kit: MKH40 (best choice for flute — RF condenser, exceptional clarity), DPA 4099 (clip-on if movement needed), or Countryman B3 (omni clip-on if available). Flute fundamentals start at C4 (262Hz). Confirm mic at soundcheck and adjust EQ accordingly.

Band	Freq	Type	Gain	Q	Notes
LC	200 Hz	LC	—	18 dB/oct	Starting point — adjust based on mic and instrument range (piccolo will want 400Hz+)
L	—	OFF	—	—	—
1	400 Hz	Bell	-2 dB	1.5	Generic mud cut starting point
2	—	OFF	—	—	Adjust based on mic
3	4 kHz	Bell	+1 dB	1.8	Generic clarity boost — adjust based on instrument
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	—	OFF	—	—	Preserve flute breath and air

Engineer Notes: PLACEHOLDER EQ — DO NOT USE AS-IS. Confirm mic selection at load-in and adjust before soundcheck. If MKH40, expect cleaner signal needing minimal EQ. If clip-on, expect more body resonance and stand noise to manage.

■ TROMBONE ■

Ch 23 | Trombone (Bone)

Mic: Warm Audio U87 Jr

Mic Notes: Warm Audio U87 Jr — LDC, U87-style, on trombone. Trombone fundamentals start around E2 (82Hz, low B-flat goes lower). For S&G; material trombone is rare — likely jazz-tinged tracks like 'Mrs. Robinson' bridge. Brass needs to sit warmly, never bark.

Band	Freq	Type	Gain	Q	Notes
LC	60 Hz	LC	—	18 dB/oct	Engaged. Trombone low B-flat = 58Hz; HPF at 60Hz preserves fundamental
L	100 Hz	Shelf	+1 dB	—	Subtle body for low brass register
1	400 Hz	Bell	-2 dB	1.4	Cut bell resonance/honkiness
2	1.5 kHz	Bell	-2 dB	1.5	Tame brass bark — most important cut on this channel
3	5 kHz	Bell	+1 dB	1.8	Bite for definition without aggression
4	—	OFF	—	—	—
H	—	OFF	—	—	No high shelf per spec
HC	10 kHz	HC	—	12 dB/oct	Engaged. Trombone has nothing useful above 10kHz

Engineer Notes: The 1.5kHz cut is the most important move — keeps trombone warm and supportive rather than aggressive. Subtle 5kHz bite gives definition. For S&G; material the bone should blend with the C422 horn pair, not stick out.

REFERENCE — Wing EQ Band Structure & Pan / Bus Routing

WING CHANNEL EQ — BAND STRUCTURE (in order, low to high):

LC — Low Cut filter (independent of EQ bands). Slopes: 6, 12, 18, 24 dB/oct.

L — Lowest EQ band. Switchable Bell/Shelf in plugin settings (default Shelf).

1, 2, 3, 4 — Four parametric bell bands, fully sweepable.

H — Highest EQ band. Switchable Bell/Shelf (default Shelf). NOT IN USE on this show per spec.

HC — High Cut filter (independent). Slopes: 6, 12 dB/oct.

SUGGESTED PAN POSITIONS:

Drums: Kick/Snare center | Hat 9-10 o'clock | OH center (mono) | Aux Perc 11 o'clock

Bass: Center

Electric Guitar: 2-3 o'clock | Acoustic Guitar: 9-10 o'clock | Keys: Center

Piano Lo: 10 o'clock | Piano Hi: 2 o'clock

Violins 1-3: 10 o'clock | Violins 4-6: 2 o'clock | Cellos: Center

Horns L (Ch 20): Hard Left | Horns R (Ch 21): Hard Right (XY pair) | Trombone: Center

Flute (Ch 22): Center

BUS GROUPING SUGGESTIONS:

Group 1 — DRUMS: Ch 1-5

Group 2 — RHYTHM SECTION: Ch 6 Bass, Ch 7 Electric Gtr, Ch 8 Acoustic Gtr, Ch 9 Keys

Group 3 — PIANO: Ch 10-11 (stereo)

Group 4 — STRINGS: Ch 12-19 (six violins + two cellos)

Group 5 — HORNS: Ch 20-23 (C422 stereo pair, flute, trombone)

SOUNDCHECK PRIORITY ORDER:

1. Confirm Ch 22 mic selection and update EQ accordingly
2. Drums (Ch 1-5) — establish kit baseline
3. Bass + Acoustic Gtr (Ch 6, Ch 8) — most critical channels for S&G material
4. Piano stereo pair (Ch 10-11) — verify pan positions match visual stage placement
5. Strings (Ch 12-19) — check feedback margin on each B3 omni at performance level
6. Horns (Ch 20-23) — verify C422 XY stereo image and pan
7. Vocals (Ch 24+) — separate document

VENUE NOTES — GREAVES SPECIFIC:

- Hardwood floor + wood acoustic shell = warm low-mid emphasis. Most channels have a 250-400Hz mud cut
- Adjustable acoustic shell — coordinate with venue if shell can be adjusted for the show
- Stage has Steinway and Baldwin 9' grand pianos — confirm which is being played
- 637 seats moderate-density audience absorption — RT60 will drop somewhat with full house
- Six B3 omnis on violins = significant gain-before-feedback challenge in this hall — monitor carefully

KSO does Simon & Garfunkel — Behringer Wing FOH EQ — Channels 1-23 — Greaves Concert Hall, NKU — Vocal/acoustic-forward style — Whole-dB values, no high shelf bands, compression separately documented.